

Chapter – 6

ROLLING MILLS

6.1 Basics of Rolling

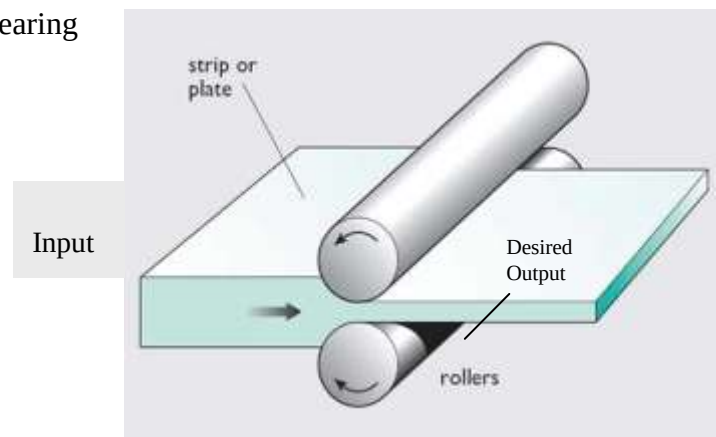
Processes of metal forming are

- | | |
|--------------------|------------------------|
| 1. Rolling | 2. Forging |
| 3. Extrusion | 4. Wire drawing |
| 5. Deep Drawing | 6. Sheet metal forming |
| 7. Stretch Forming | 8. Foundry |
| 9. Bending | 10. Shearing |

Basic Definitions

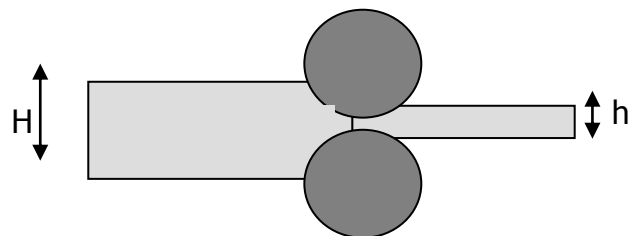
Rolling

Rolling is plastic deformation of the Metal by passing between rolls to give it the desired shape.



Draft

Difference in height or thickness of the material before rolling (**H**) and height or thickness after rolling (**h**) is called draft (**H-h**). It indicates how much the metal has been pressed during rolling.



Spread

Difference in width of material after rolling (**b**) and width before rolling (**B**) is called spread (**b-B**). It indicates how much the metal has spread during rolling.

Elongation

Difference in length of material after rolling (**l**) and length before rolling (**L**) is called elongation (**l-L**). It indicates the increase in length during rolling.

Reduction

Difference in area before rolling (**A**) and area after rolling (**a**) is reduction (**A-a**). It indicates how much the cross section area has been reduced during rolling.

Coefficient of Reduction

Ratio of area before rolling (**A**) and area after rolling (**a**) is called coefficient of reduction (**A/a**). It indicates how many times the area has been reduced during rolling.

Rolling Constant principle

It states the volume of material will remain same before and after rolling. It is useful in finding input and output sizes.

Basic terminology used for measuring mill efficiency

Mill Availability

It indicates the availability of the mill for rolling. In this the planned shutdown and capital repairs are subtracted from the total calendar hours.

$$\text{Mill Availability} = \frac{\text{Calendar Hours} - (\text{Capital Repairs} + \text{shutdown})}{\text{Calendar Hours}} \times 100$$

Calendar hours in a year means 24 x 365 days (or 366 in case of leap year)

Mill Availability is expressed in percentage

Mill Utilization

It indicates the utilisation of available mill for rolling. In this the planned delays subtracted from the available hours.

$$\text{Mill Utilization} = \frac{\text{Calendar Hours} - (\text{Planned Repair} + \text{Total Delays})}{\text{Available Hour}} \times 100$$

Available hours = Calendar Hours - (Planned Shutdown + Capital Repairs)

Mill utilization is expressed in percentage

It is a measure of effective utilization of time available

Hot Hours

Hours during a day or month or year during which rolling actually took place. Its unit is hours.

Yield

It is the ratio of useful output to input expressed as percentage $\text{Yield} = \frac{\text{Output}}{\text{Input}} \times 100$

It is a measure of efficient utilization of input and is expressed in percentage.

Rolling Rate

It is tonnage rolled in an hour. It is a measure of speed of rolling and its unit is tons/hour.

Hot and Cold Rolling

Hot Rolling:. Hot rolling is a metalworking process in which metal is heated above the recrystallization temperature to plastically deform it in the working or rolling operation. This process is used to create shapes with the desired geometrical dimensions and material properties while maintaining the same volume of metal. The hot metal is passed between two rolls to flatten it, lengthen it, reduce the cross-sectional area and obtain a uniform thickness. Hot-rolled steel is the most common product of the hot rolling process, and is widely used in the metal industry either as an end product or as raw material for subsequent operations. Hot rolling breaks up the existing grain structure and promotes recrystallization, leading to the formation of new grains with strong boundaries and a more uniform grain structure.

Hot rolling improves:

- Toughness and strength
- Ductility
- Resistance to vibration and shock
- Formability
- Weldability

Hot-rolled steel products are classified into four groups:

- Flat
- Long
- Seamless
- Specialty

Typical applications of hot-rolled steel are:

- Automotive structural parts such as frames
- Tubular products such as pipe and gas cylinders
- Machine structures such as saws and springs
- Agriculture equipment
- Metal buildings
- Guard rails

Cold Rolling: Metal forming process in which the shape and structure of steel is altered through drawing, extruding, hammering, pressing, rolling, spinning, and/or stretching at temperatures below the steel's recrystallization point (usually room temperature).

Cold rolling is a technique where a metal strip or sheet is passed between two rollers and then squeezed and compressed. The level of strain present determines the properties and hardness of the finished material.

This process is widely used for surface finish and high-quality dimensional accuracy, which can help prevent material damage and corrosion

The typical applications of cold rolling are in the production of:

- Filing cabinets
- Metal furniture
- Computer cabinets
- Water heaters
- Exhaust pipes
- Steel drums

Through cold rolling of steel, the strength and hardness of the material is enhanced significantly, which in turn improves the corrosion resistance of metal in many ways. This is brought about by the addition of more tension and energy to the product.

Recrystallization temperature is the temperature in rolling above which we get strain free grains and minimum residual stresses in rolled metal. It is normally 0.5 to 0.7 times of melting point of the metal.

The difference between the two types of rolling is the temperature at which they are processed. Hot Roll is processed above the re-crystallization temperature. Its grains reform after the rolling process and it is left in a stress-free state. Cold Roll is processed below the re-crystallization temperature. Its grains remain flattened and elongated, leaving the material in an anisotropic state and full of cold work.

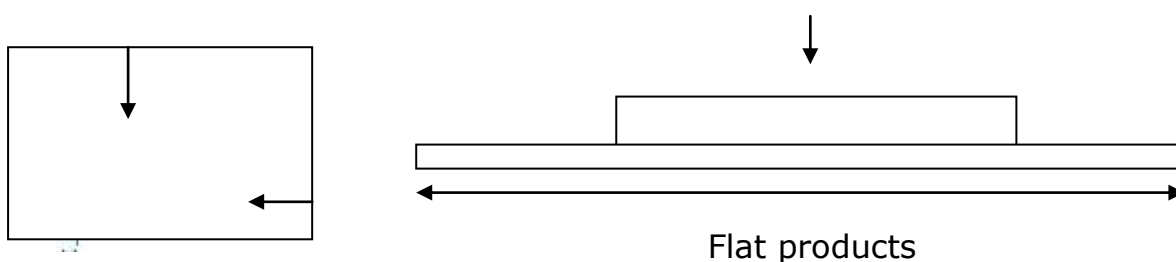
FACTORS	HOT ROLLED CHARACTERISTICS	COLD ROLLED CHARACTERISTICS
Layout Orientation	The material characteristics is identical in all directions	Care must be exercised in layout. The grains of the material are deformed during the rolling process and stay deformed. The material will be stronger with the grain than against the grain.
Price	Less Expensive	More Expensive
Strength	Weaker	Stronger
Weldability	Excellent for welding	Weldable, but the material will take on the properties of hot roll wherever it is welded
Machinability	Experiences no warping when machined	The removal of too many residual stresses during machining will throw the material out of equilibrium and cause deflection and warping
Dimensional tolerances of stock	Fair. Deviations from the stated size are present due to surface scale and thermal shrinkage	Good. It is not as accurate as ground stock, but better than Hot Roll.
Surface Finish	Fair to Poor The surface of the material will be covered with carbon scale	Good. Not as good as ground stock, but much better than Hot Roll.

All SAIL integrated plants have hot rolling mills whereas only Bokaro, Rourkela and Salem have cold rolling mills.

Long and Flat Products

During rolling, when the metal is pressed from both the top–bottom and from the sides (perpendicular directions), the material is forced to flow mainly along its length to maintain constant volume. This is called **Long Product Rolling**, where the length increases while the cross-section is controlled. It Produces products like TMT Bars, Rebars, Angle, Channel, Joist, Rails.

When the metal is compressed only from the top and bottom by the rolls, the thickness decreases and the material spreads sideways as well as along the length. This process is called **Flat Product Rolling**, which produces sheets, plates, and strips.



Long products

In SAIL, integrated steel plants Bokaro and Rourkela produce flat products, while Burnpur is a long product plant. Bhilai and Durgapur produce both long and flat products.

6.2 Products of Rolling Mills of SAIL

Bhilai Steel Plant

- Semis (Blooms, Billets, Slabs and Narrow width slabs)
- Rails
- Heavy Structurals (Beams, Channels, Angles, Crossing Sleepers)
- Merchant Products (Angles, Channels, Rounds and TMT Bars)
- Wire Rods (TMT, Plain and Ribbed)
- Plates

Bokaro Steel Plant

- HR coils, sheets, plates
- CR coils, sheets
- Galvanized plain and corrugated sheets

Durgapur Steel Plant

- Semis (Blooms, Billets, Rounds)
- Merchant Products (TMT Bars, Rebars)
- Medium Structurals (Beam, Joists, Channels, Angles)

- Wheel and axle

Rourkela Steel Plant

- Semis(slabs)
- Plate Mill Plates, Special plates
- HR Plates, Coils
- CR Coils and Sheets
- Galvanized plain and corrugated sheets
- Silicon Steel Sheet & coils
- HFW Pipes(ERW) & SW Pipes

IISCO Steel Plant, Burnpur

- Structurals(Beam, Channels, Angles)
- TMT Bars, Wire Rod Coils Products

Alloy Steel Plant, Durgapur

- Alloy and Stainless steel Slabs, Blooms, Billets, Bars, Plates
- Stainless and Hadfield Manganese steel plates

Salem Steel Plant

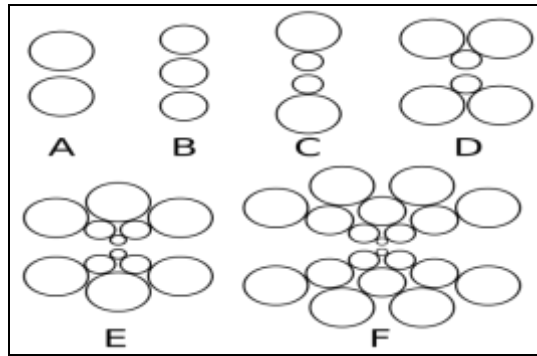
- Hot rolled carbon and Stainless-steel flat products
- Cold rolled stainless steel sheets and coils

Visvesvaraya Iron and Steel Plant, Bhadrawati

- Semis, Bars

Different configuration of rolls used in rolling mills

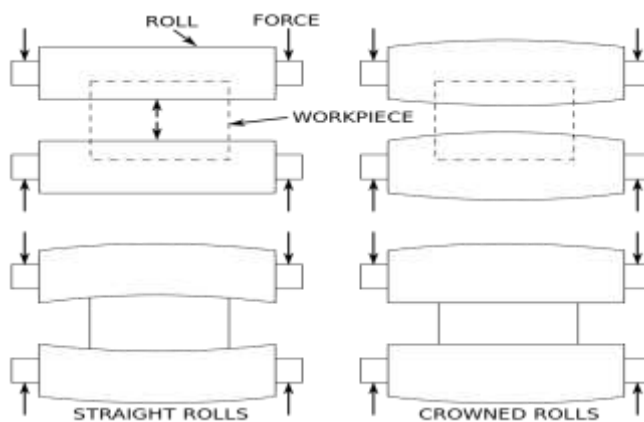
Mills are designed in different types of configurations, with the most basic being a two-high non-reversing, which means there are two rolls that only turn in one direction. The two-high reversing mill has rolls that can rotate in both directions, but the disadvantage is that the rolls must be stopped, reversed, and then brought back up to rolling speed between each pass. To resolve this, the three-high mill was invented, which uses three rolls where two consecutive rolls rotate in opposite direction; the metal is fed through two of the rolls and then returned through the other pair. The disadvantage to this system is the workpiece must be lifted and lowered using an elevator. All of these mills are usually used for primary rolling and the roll diameters range from 60 to 140 cm (24 to 55 in).



Various rolling configurations. Key: A. 2-high B. 3-high C. 4-high D. 6-high E. 12-high cluster & F. 20-high [Sendzimir Mill](#) cluster

To minimize the roll diameter a four-high or cluster mill is used. A small roll diameter is advantageous because less roll is in contact with the material, which results in a lower force and power requirement. The problem with a small roll is a reduction of stiffness, which is overcome using backup rolls. These backup rolls are larger and contact the back side of the smaller rolls. A four-high mill has four rolls, two small and two large. A cluster mill has more than 4 rolls, usually in three tiers. These types of mills are commonly used to hot roll wide plates, most cold rolling applications, and to roll foils.

Roll deflection during rolling



6.3 Applications of Rolled Products of SAIL

Hot Rolled Coils, Sheets

Used for construction of tanks, railway cars, bicycle frames, ships, engineering, military equipment, LPG cylinder, automobile and truck wheels, frames, and body parts. HR coils are also used as feedstock for pipe plants and cold rolling mills where they undergo further processing. Hot Rolled Chequered coils and plates are being produced for using in floor as anti-skidding.

Plates

Steel plates are used mainly for the manufacture of bridges, dams & windmills, steel structures, ships, large diameter pipes, storage tanks, boilers, railway wagons, and pressure vessels. SAIL also produces weather-proof steel plates for the construction of railcars. SAIL is one of the major producers of wide and heavy plate products in India.

Cold Rolled Products

The products of the cold rolling mills include cold rolled sheets and coils, which are used primarily for precision tubes, containers, bicycles, furniture, whitegoods industry and for use by the automobile industry to produce car body panels. Cold rolled products are also used for further processing, including for color coating, galvanizing and tinning. Galvanized Sheets are used in roofing, paneling, industrial sheeting; air condition ducting and structural applications. Electrolytic Tin Plates are used in containers for packaging of various products including edible oils, Cola, Fruit Juices, Pickles and confectionary items.

Railway Products

Rails are one of the main rolled products by SAIL. It is used primarily to upgrade and expand railway networks.

Structurals

I-beams, channels, and angle steel are used in mining, construction of tunnels, factory structures, transmission towers, bridges, ships, railways, and other infrastructure projects.

Bars and Rods

Reinforcement steel and wire rods are primarily used by the construction industry.

Semi-Finished Products

Semi-finished products (blooms, billets and slabs) are converted into finished products in SAIL's processing plant and, to a lesser extent, sold to Re-rollers for conversion to finished products.

Alloy and Stainless Products

Alloy and special steel products with alloyed elements including chromium, nickel, vanadium and molybdenum are primarily used for sophisticated applications, including in the automobile, railway, aerospace, power, nuclear, submarines and defence industries. Special alloy steel billets and bars made for defence are used in shell making. Jackal and spade plates are critically applied for armour and ammunition vehicle application only. Corrosion resistance cold rolled stainless steel coils and sheets are used for diverse applications including household utensils, automobile trims, elevators, fuel, chemical, fertiliser, LPG tanks, atomic power, boilers, heavy engineering, dairy and food processing equipment, coin blanks, building and interior decoration, and pharmaceutical equipment.

Speciality Products

Speciality products include electrical sheets, tin plates, and pipes. Electrical sheets are cold rolled products of silicon steel for electrical machinery. Pipes are longitudinally or spirally welded from hot rolled coils for transporting water, oil, slurry and gas.

6.4 Hot Rolling

The rolling process which is done above recrystallization temperature is called hot rolling. More reductions are achieved in hot rolling as compared to cold rolling.

6.5 Reheating Furnaces

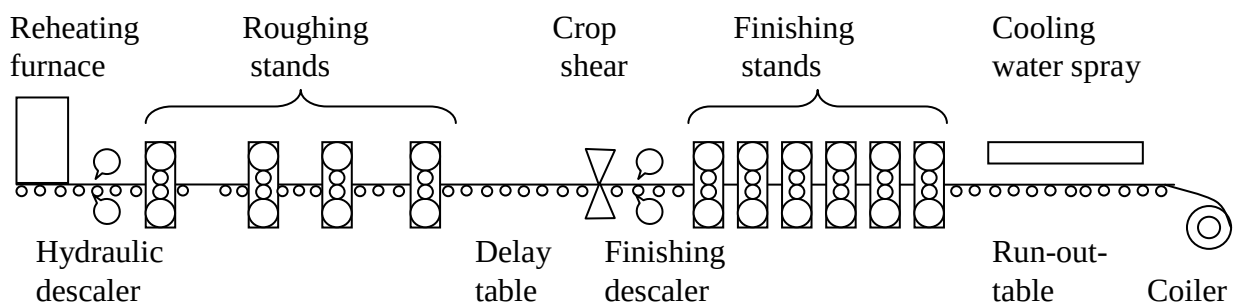
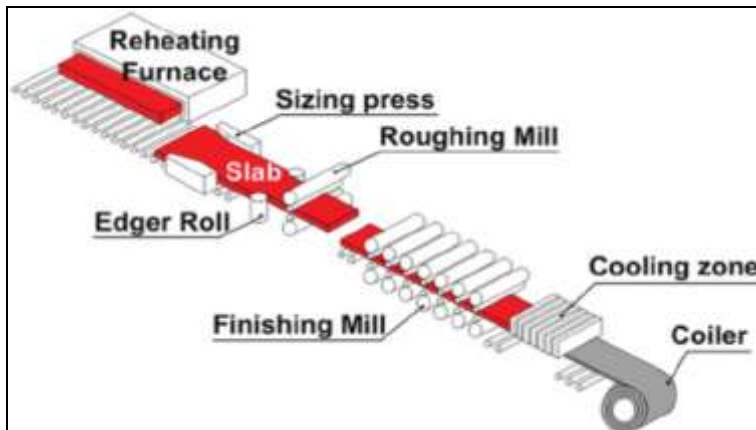
In the reheating furnaces the input materials are heated to a specified temperature and soaked for given time depending upon size of input slab and their metallurgical requirements for which it is planned to be rolled. Ideally, it is aimed to equalize the surface and the core temperatures of the slab. Well-soaked slabs are discharged from the furnace at dropout temperature of 1100-1300°C. The furnace discharge temperature is also to compensate heat losses in downstream operation and thereby it depends on it.

Types of Reheating Furnaces:

In Primary mills Soaking pits are used which are primarily batch type furnaces and at present not in operation. In secondary (finishing) Mills, continuous reheating furnaces are used. Continuous reheating furnaces are mainly of two types: pusher type and walking Beam type. These furnaces mainly have firing system with mixed gas (mixture of coke oven gas and blast furnace gas) which are readily available in Integrated Steel plants.

6.6 Rolling of Flat Products

Layout of a typical Hot Strip mill is shown in figure below:



Process installations shown in the figure above and operations performed are briefly described below:

PROCESS:

Reheating

Walking Beam Reheating Furnaces are having 6 zones (Preheating zone top & bottom, Heating zone top & bottom and Soaking zone top & bottom). Preheating, heating and soaking zones have firing system with mixed gas, Slabs are heated, soaked to have uniform temperature across the cross section of the slabs and discharged at 1100-1300°C based on the grade of the steel and the planned dimension.

Descaling:

Layers of scales (oxides of iron) are formed on the surface of the slab during its heating inside the furnaces. The scales are removed by using high pressure water jet. The descaling unit consists of headers fitted with nozzles for spraying water both at top and bottom surface. Descaling is a very important precondition for rolling for defect free products.

Roughing:

The major reduction to the input material is given at Roughing mill group to get the desired thickness for Finishing Mill group. Main reduction in thickness is achieved at Roughing mills and comparatively smaller draft is given at Finishing mills. . For example, if strip of 2 mm thickness is to be produced from 220 mm thick slab, typically, thickness will be reduced from 220 mm to 26-40 mm at Roughing stands (continuous or reversible) based on the strip width .Final drafts are given at Finishing mill to get the planned dimensions.

Finishing:

Final required dimensions of the end products are achieved in finishing process. Finishing temperature of the strip(which is temperature at the last finishing stand) is a critical parameter, is maintained and not being allowed to decrease below a specified temperature for particular grades of steel . This is done to achieve the metallurgical properties.

Cooling

Before the hot rolled strips are coiled in the coiler, they are cooled at specified cooling rate on the run-out-table to achieve the desired coiling temperature. It is very much important for getting the desired metallurgical properties of the strip. The number of the water banks to be operated is decided by the targeted cooling rate of the strip to achieve the required coiling temperature.

Coiling:

The Strip moves over ROT and gets coiled in coilers. The coils are taken out of the coilers, strapped on the body and marked for identification.

Coil Finishing & Dispatch

Coils are sent for further processing or directly sold as hot rolled products. Weighment of the coils, inspection and sampling for both chemical and physical testing is done before processing and dispatch.

Equipments:**Reheating Furnace:**

Re-heating furnaces of hot Strip mill are mainly of two types- pusher type or walking beam type. The walking beams are hydraulically operated and the movement of slabs inside the furnace is carried out by a set of moving & stationary beams. In pusher type furnaces the movement is achieved by pushing the slabs one after another. The combustion system is mainly of recuperative type with heating from top and bottom. Facilities of skewed skid system are there in the furnace to compensate for temperature variations.

Descaler:

The surface of slab moving through water hydraulic descaler and the rolling bar/strip is impinged by high-pressure water jets. At many places, a mill stand with a pair of vertical rolls, called vertical scale breaker, may also be provided to remove the scale from edges of the slab.

Roughing stand:

Roughing mills are generally having one stand, two stands and multi-stands with 4-high configuration. These stands can be reversing, non-reversing or combination types. Universal type roughing stands are equipped with vertical edgers for controlling spread of material in lateral direction. Rolls of roughing stands are cylindrical and are cooled simply by water.

Finishing Stands:

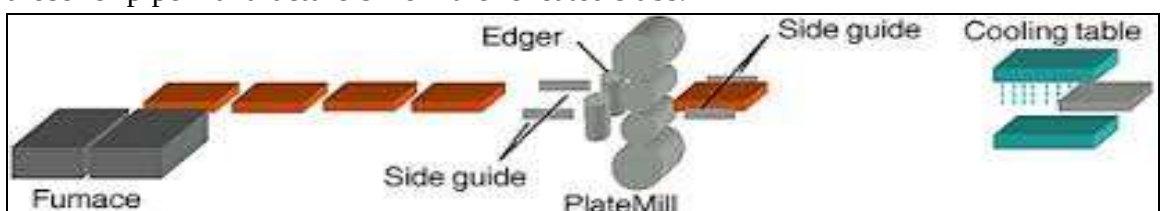
Finishing Stands of Hot strip mill normally have 5-7 stands in 4-high construction (combination of work rolls and back up rolls). These stands are in tandem and strip passes through them on continuous basis. Roll force, roll gap can be controlled through different mechanisms such as roll bending, pair crossing, roll shifting and varying crown of rolls. Cooling of the rolls is achieved by spraying water/water oil mixture over backup and work rolls.

Coilers:

The coilers are used for coiling the strips mainly with the help of Pinch rolls, wrapper rolls and mandrel.

Plate Mill

The modern plate mill of both RSP and BSP rolls out heavy and medium plate as well as those for pipe manufacturers from the reheated slabs.



MAJOR Parameters and factors affecting rolled products and their control:

The following major factors affect the quality of rolled products

a) Temperature:

The desired temperature at various stages of rolling needs to be maintained for attaining product within dimensional tolerances and properties.

b) Roll conditions:

Roll change schedule has to be strictly adhered to and roll cooling conditions need to be monitored continuously. Shape and dimensional tolerances also depend on the above mentioned conditions.

6.7 Rolling Of Long Products

Long products have to be hot rolled only, to facilitate the large reduction to be made in passes. The mills can be basically classified into primary mills and secondary mills. Primary long product mills manufacture semis mainly blooms and billets. The long product mills which produce finished products like beams, angles, channels, bars, wires and rods, TMT bars and rails, are called finishing mills

Rail Mills

Rails are produced either using the two-high rolling method on two-high reversing mills or on three-high mills or, increasingly today, using the universal rolling method. The universal rolling method has proved superior to the conventional methods due to the following advantages: Closer rail tolerances, better surface quality and less roll wear.



Rail mill

Structural Rolling Mill

Rolling mills having facilities for rolling the billets, blooms, slabs and beam blanks. Input materials are heated in reheating furnaces (Walking Beam or Pusher Type) to desired temperature (1100 to 1300°C). The Rolling is done through Rolling Stands (Reversing or Continuous) to achieve the desired shape and dimensions. Brand marking on finished product is done at final stands.



Rolling of TMT Bars/Rebars (In Merchant Mill DSP)

TMT Bars are produced using a special rolling and cooling process called **Thermo-Mechanical Treatment (TMT)**. The process combines **rolling, rapid cooling (Quenching), and self-tempering** to obtain a strong outer layer and a ductile core.

Wire Rod Mill

The objective of a **wire rod mill (WRM)** is to roll steel billets into **wire rods**. The production volume of **wire rods** in WRM is subject to the size of product to be rolled and mill availability. High productivity is achieved in case of thicker size rolling.



PROCESS

The processes of rolling of long products can be basically classified into following heads.

Reheating:

The reheating of inputs is done to make the material plastically deformable and pliable for rolling to give the desired shape and size.

Roughing:

Roughing is done to give the input a rough shape. The maximum reduction in cross section is given in roughing mills.

Intermediate Rolling:

Intermediate rolling constitutes taking the roughing mill output as its input. Output of intermediate rolling is sent to finishing mills.

Pre-Finishing & Finishing rolling:

In Pre-finishing & Finishing mills the finished profile shape is made. Pre-finishing Mills It takes metal from intermediate stands as its input. The reduction given in these stands is lesser as compared to roughing and intermediate mill. Finished shape is achieved in finishing mills. Finishing mills is critical as the final shape of the material is made in these stands, Also, the branding is done at the last finishing stands. In case of TMT bars thermo-mechanical

treatment of bars is done after finishing mill. In this the red hot finished TMT bars enter a **Quenching** box where high-pressure water rapidly cools the **outer surface** of the material, due to which the outer layer becomes cold and hard whereas the inner part remains hot and soft.

Cutting and stamping:

The finished bar is cut to desired lengths as per customer requirement.

In secondary mills stamping of cast number and other details are done on the rolled products in hot condition. These are required for identification and traceability of product and correlation with test results of destructive tests.

Finishing:

The finishing is done after the bar has been cooled to ambient temperature. Finishing activities at different mills may involve all or few of the following steps:

- Straightening either by roller straightening machines or by pulling the ends
- End finishing either by milling or cold cutting
- Online non-destructive testing of defects
- Heat treatment

Inspection:

Inspection is carried out by the producer and/or by customer deputed agency and/or by third party to inspect the products and ensure no defective products are sent to customer. Depending on the specifications, customer requirement the inspection may be for all or few of the parameters like Dimension, Straightness, Squareness, Surface Quality, Branding, colour coding and stamping are done.

Dispatch:

The products are sent to required destinations primarily by rail and in some cases by road. The activities involve documentation (preparation of dispatch advice (DA), test certificate (TC) and clearance by train examiner (TXR) in case of rail dispatches. In some cases packaging is done prior to dispatch.

Equipments

Reheating furnaces:

The heating of inputs is done in reheating furnaces.

Primary long product mills use batch type furnaces (Soaking pits) for reheating of ingots. Finishing mills use continuous furnaces (either pusher type or walking beam type) for reheating of inputs.

Stands:

Equipments in which rolling is done are called stands. They may consist of all or few of the following components – Rolls, housings, bearings, chocks, couplings with drives, manipulators, tilters, screw down mechanisms. The stand may have horizontal rolls or vertical rolls or combination of both types of rolls.

Accessories of stands:

Accessories of stands consist of mainly roll cooling arrangements, guards, guides, tackles and grease systems etc.

Automation:

Most of the rolling mills have Level-II automation and controlled with PLC for achieving best output and reduction in down time.

Drives:

In most of the mills reversible electrical drives of high ratings are required to drive the rolls. In certain cases the drives give their output directly to rolls through spindles.

Shears / Cutting Saws:

Shears are used to cut sections (Blooms/Billets) in primary mills. Cutting saws are used to cut products of finishing mill to desired lengths, cut crops and samples.

Straightening machines:

Two types of straightening machines are in use in finishing mills. Roller type in which the products are straightened by alternately bending the products in opposite directions between rotating rollers. In case of lighter profiles the straightening is done by pulling from both the ends.

End finishing equipments:

Ends with square cut and good surface finish required in some finished products is achieved by milling or cold cutting with carbide saws.

Online testing equipment:

In some finished products online non-destructive testing is done by ultrasonic testing machine (for inside defects), X- ray for subsurface defects and eddy current testing machine (for surface defects)

Auxiliaries:

Auxiliaries such as cranes, roll tables, material handling equipments etc. are very important for integrated functioning of mill.

Defects of hot rolled long products

Defects due to which hot rolled long products are rejected may be broadly classified as:

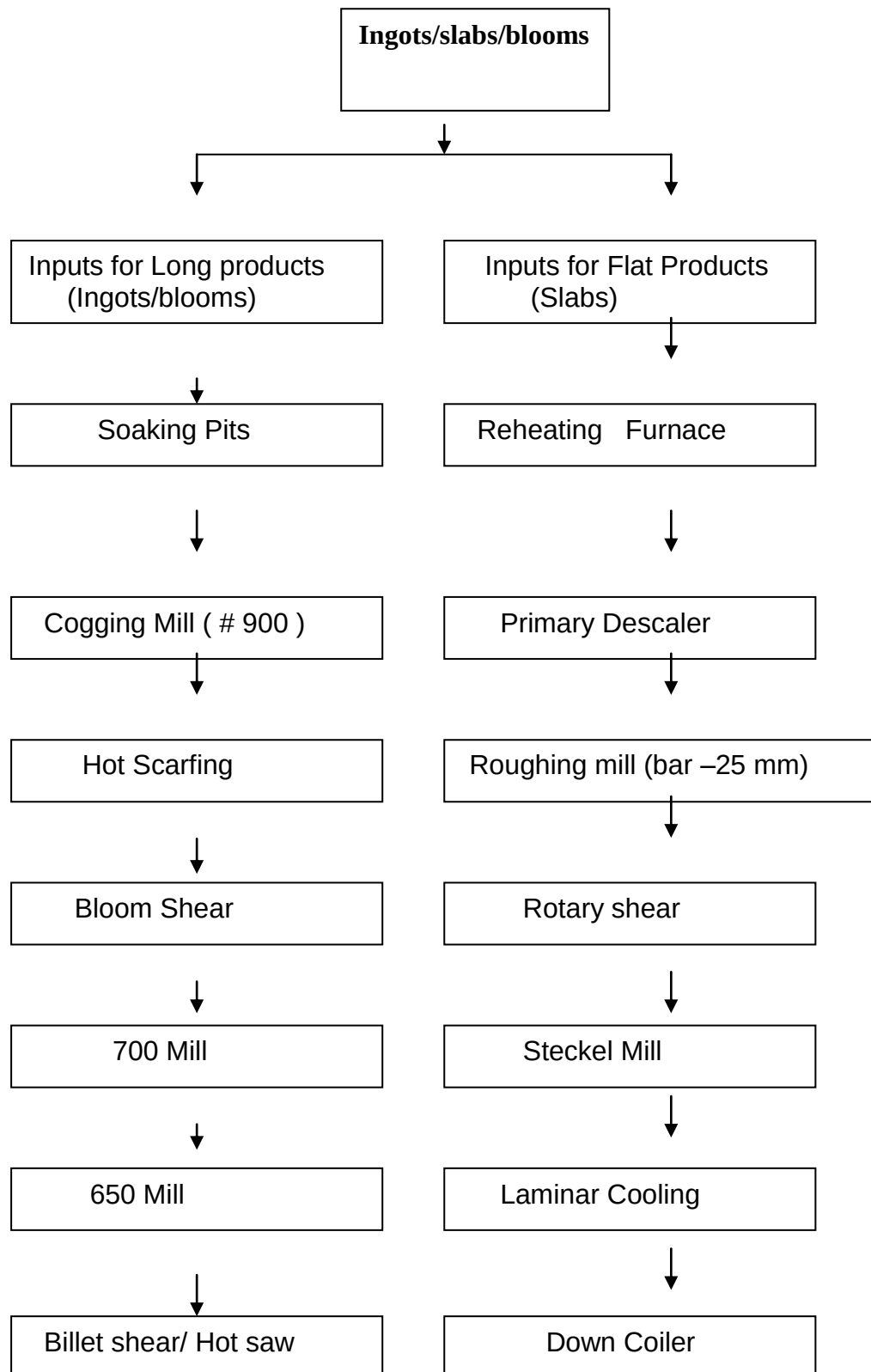
- a) **Rolling defects:** Defects induced during the rolling process are classified as rolling defects.
- b) **Steel defects:** Defects resulting from steel making practices and getting carried forward during the process of rolling to the end product may be classified as steel defects.

Rolling defects:

Some defects of hot rolling are:

Fins and overfills, underfills, Slivers, Laps, Fire cracks and roll marks, Rolled-in scale, Buckle and kink, Camber, Twist, Shear distortion, Out of square, Burnt edges, ridge-buckle, wedge.

Rolling of Special Steels\ Process Flow



Special steels including Alloy and Stainless have both continuous cast and Ingot teeming products. Grades having high carbon and high alloy are hot transferred. All ingots and continuous cast slab or blooms are charged into soaking pit furnaces.

The soaked material is transferred from soaking pit to cogging mill by cranes. The scarfed bloom is taken to bloom shear for discarding hot top and bottom end. The sheared bloom is ready for finishing rolling either round or square sections.

The rolled bloom or billet is cut to length by shear or hot saw depending on the final requirements. The dispositioning depends on the grades and final property requirement.

The stainless steel slabs received from Alloy Steel Plant, are heated in a walking beam furnace in Salem Steel Plant. After this the slabs are transferred to steckel mill for hot rolling.

Slabs are rolled in 4 high roughing mills to get required T-Bar thickness and coiled in the down coiler.

Inspection

Rolled bars, Billets and Blooms are visually inspected in conditioning shops after exposing the surface either by grinding (zig-zag or ring), pickling or shot blasting. After exposing the surface, defects are removed by grinding to the allowable depth. Defective portions are discarded by gas or saw cutting. Ultrasonic flaw detectors are used for inspection of Rail & pipes.

Surface inspection and dimensional check of hot rolled coils are done before coiling and disposition.

Testing

Samples are collected from rolled products for testing for both physical and chemical. Different tests are carried out for checking internal defects, Grain size, Microstructure, Inclusion, Hardness, upsetting, toughness etc.

Input hot rolled coils testing is done as they are taken up for cold rolling.

Heat treatment for Long Products

This is done at the finished stage of processing depending on mechanical property requirement.

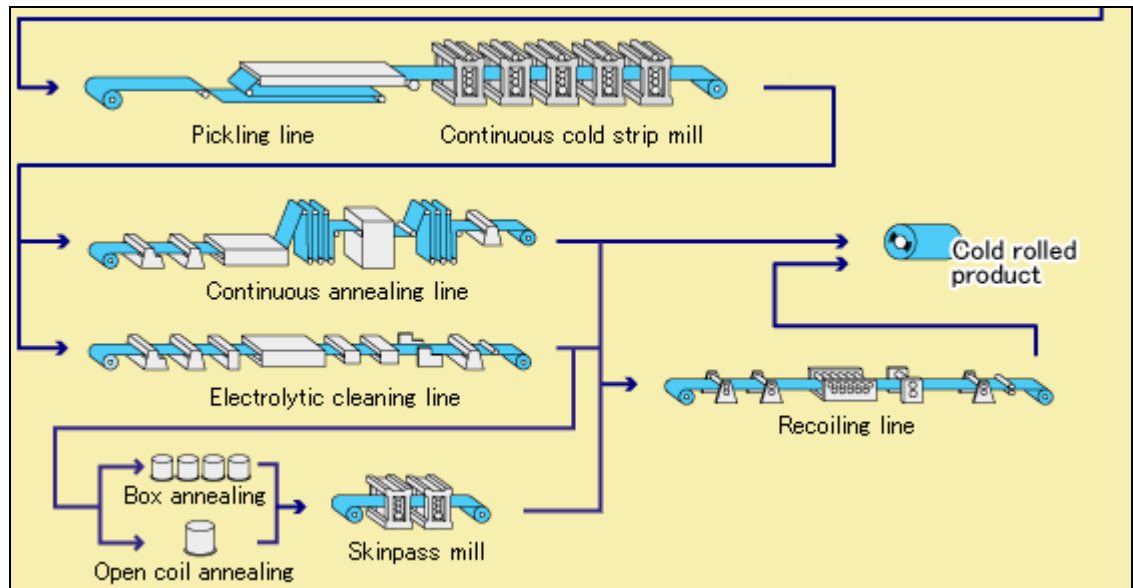
SPECIAL PLATE PLANT (SPP)

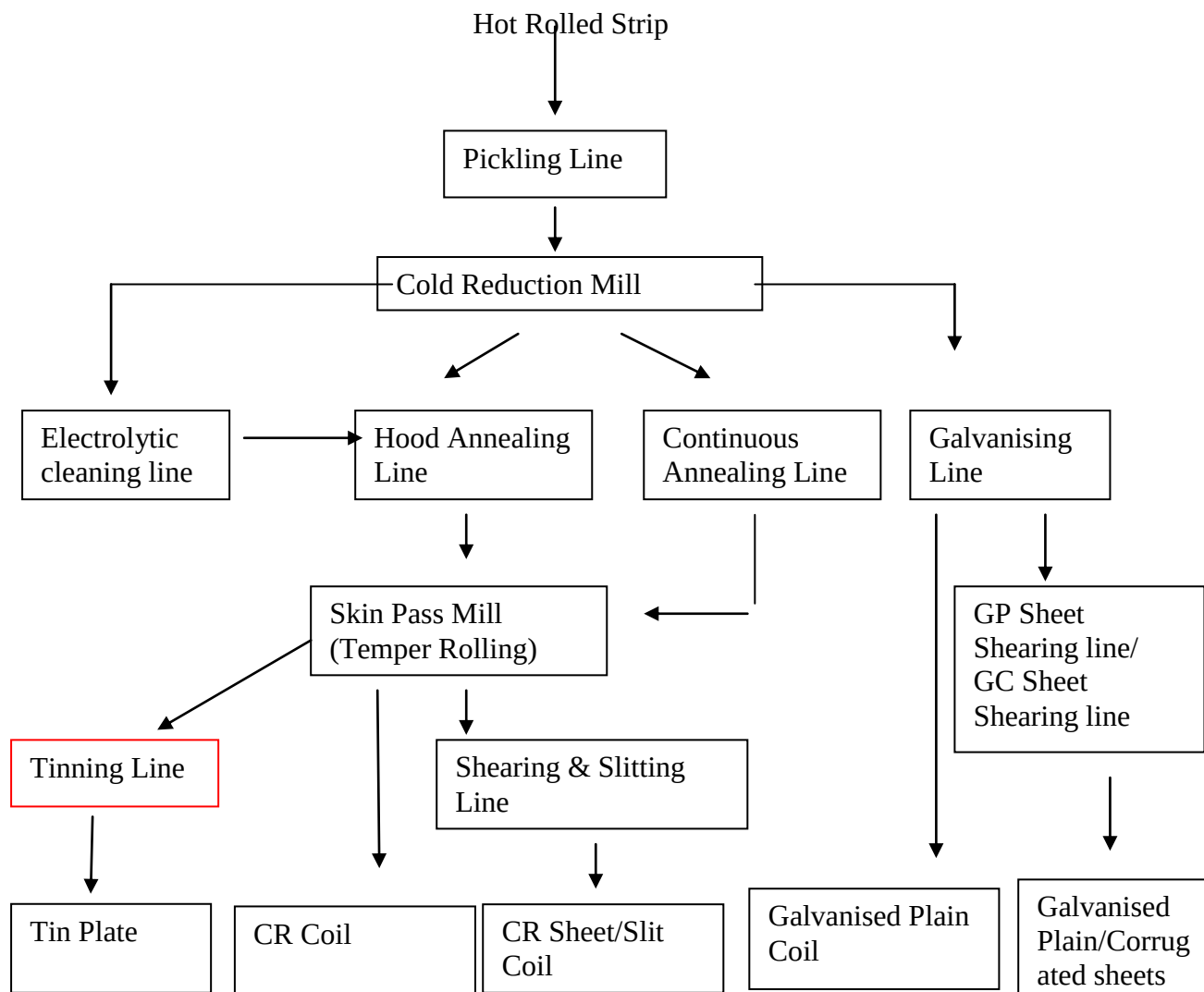
Special Plate Plant (SPP) of Rourkela Steel Plant caters to the needs of Defence & Space requirements. Special Plate Plants is **the only unit in India** producing various grades of quenched and tempered special steel plates, Armour Plates & Components for Defence in larger dimensions. All these steels are weldable.

6.8 Cold Rolling

COLD ROLLING MILL (CRM)

Purpose of cold reduction is to achieve, a reduction in the thickness, a desired surface finish, desirable mechanical properties, close dimensional tolerance and producing as per customer requirements. These thickness reductions are achieved through multi-pass rolling in a reversing mill or tandem mill. Apart from such mills, a cold rolling mill complex may include other facilities for pre-and post-rolling operations. The sequence of operation and material flow in a typical cold rolling mill complex is shown in Figure 3.1.





(Fig.3.1: Typical material flow in a Cold Rolling Mill Complex)

The input to Cold Rolling Mills is the Hot Rolled Coils (HR Coils) from HSM.

N.B. –at present Tin Plate is not being produced

Pickling Lines:

During the hot rolling process, a layer of scale (Iron oxides) is formed on the strip surface, which must be removed prior to further processing. This removal of scale is performed by physically breaking of scales by mechanical means & then chemically treating the surface of hot rolled strip with an acid. The process, called ‘Pickling’, removes the remaining scale by dissolving it in acid. Hydrochloric and Sulphuric acids are most commonly used for pickling. Pickling rate with hydrochloric acid is 2.5 to 3 times higher than with sulphuric acid under equivalent bath concentration and temperature conditions.

Cold Reduction:

After pickling, the main cold rolling operation, i.e. cold reduction, is performed in cold reduction mill where pickled strip is fed between very hard rolls. Cold rolling is done

- Either in a single reversing stand, equipped with an uncoiler and a coiler, by making several passes in reversing directions;

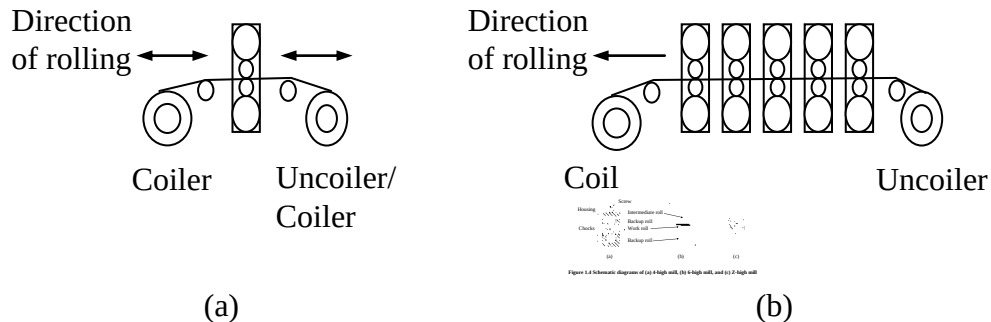
or

- In a continuous tandem mill where the strip is engaged in several stands simultaneously, enabling high-tension force to be applied.

Cold rolling in multi-stand tandem mill is widely used because of high speed of operation. The roll arrangement in each stand is 4-high, 6-high and even 20-high. The 20-high mills are used for rolling of stainless steels. Application of coolant with lubricant reduces the friction and heat generation at the roll bite and thus reduces the roll and strip temperature during rolling.

Cold Reversing Mill:

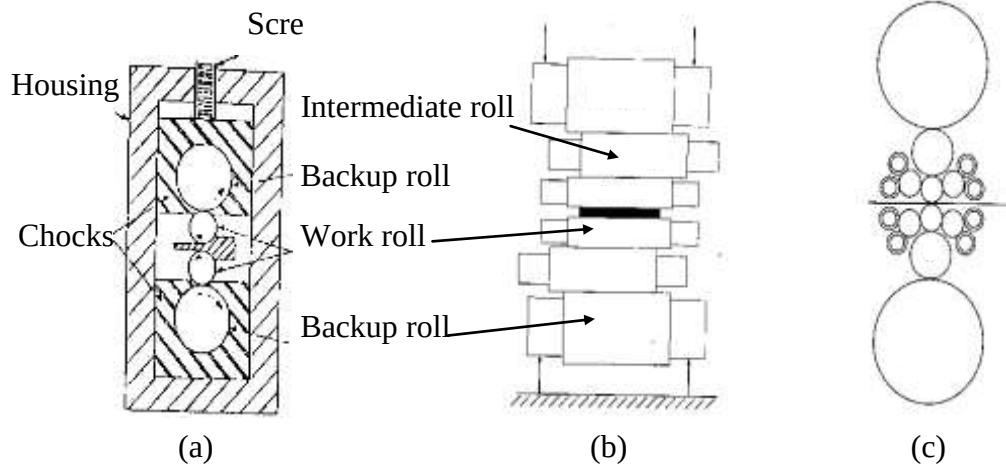
This is a 4-Hi reversing mill which makes 2-5 passes to reduce thickness. It consists of a single stand with reels located on either side of the mill. Steel strip is passed back and forth till the required thickness is obtained.



Schematic of (a) Reversing Mill and (b) Tandem Mill

Tandem Mill(TM)

In Tandem rolling, the material to be rolled undergoes reduction in all the mill stands at a time.



Schematic diagrams of 3.2(a) 4-high mill, (b) 6-high mill, and (c) Z-high mill

Each stand of tandem or reversing mills consists of a set of independently driven pair of rolls, which come in direct contact with the strip and create a converging gap for imparting deformation to the strip. These rolls are called work rolls. Comparatively larger diameter backup rolls support these work rolls. When the mill is having one pair of work rolls and a pair of backup rolls it is called 4-high mill. To impart further rigidity, in some of the mills each work roll is supported by one additional roll (intermediate roll) between the work roll and backup roll. This type of the mill is called 6-high mill. A mill in which each work roll is surrounded by a cluster of backup and intermediate rolls is called Z-high mill or Sendzimir Mill. Schematic diagrams of these mills are shown in figure 3.2.

ROLL SHOP

All Integrated Steel Plants have Roll Shops/Roll Turning Shop , which does grinding/finishing of Rolls.

These depts supplies Work roll & Back up rolls to all mills i.e. Hot strip mill, Plate Mill, New Plate Mill, Colld Rolling Mills, Silicon Mill, Pipe Plants., URM, Wire Rod Mill.

Quality of Finished product of all Mills is dependent upon quality of finishing of Rolls.

All rolls are changed after certain Tonnage rolled which is different for all mills.

ELECTROLYTIC CLEANING LINE

Electrolytic Cleaning is required in case material rolled with high percentage of oil while reduction in mills goes for annealing in furnace.

Annealing Process:

Cold rolled strip as such is not suitable for drawing and deep drawing operations due to lack of ductility. The work hardening effects of cold reduction cause the loss.

Now these CR coils are to be annealed in protective atmosphere so as:

1. To improve the mechanical properties.
2. To increase ductility, particularly to restore the normal conditions of steel after cold working.
3. To relieve the internal stresses.
4. To remove chemical non-uniformity.
5. To change the micro-structure of steel from the distorted structure of cold worked steel to the equi-axed structure.

Annealing is done in either of the following two lines:

1. Hood (Batch or Box) Annealing Line (HAL)
2. Continuous Annealing Line (CAL)

Whichever method of annealing is used, the steel is maintained under a protective (non-oxidizing) atmosphere using **hydrogen and nitrogen** to prevent oxidation of steel while it is at high temperature. For cleaner and brighter coils sometimes, Hydrogen is used as protective atmosphere though the process is costly.

Hood Annealing/ Batch annealing, Box/ Batch annealing is still the most common and convenient method of annealing and a major portion of cold rolled coils are annealed in Hood Annealing furnaces in spite of development of continuous and open coil annealing. The main reason for its wide use is that **wide range of annealing cycles can be adapted** to suit to Customers' requirements.

Different annealing cycles are followed for different grade & thickness of cold rolled coils.

Continuous Annealing:

Continuous annealing involves passing the steel through a high temperature furnace in the form of a continuous strip.

This is a much faster process compared to Hood annealing Process.

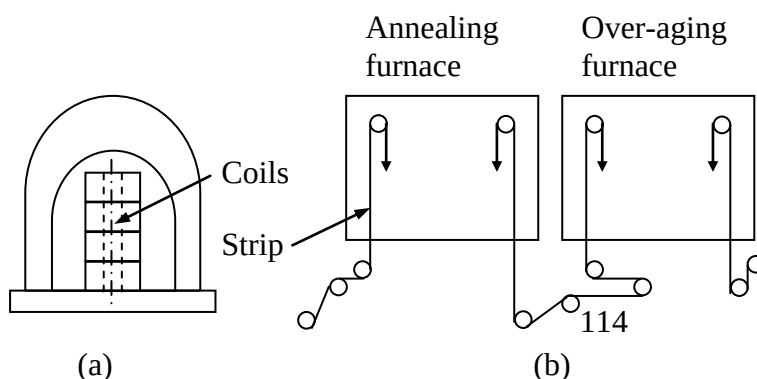


Figure: Schematics of (a) Batch Annealing and (b) Continuous Annealing

Skin Passing:

After annealing, the coils are given a further light rolling without strip lubrication. This operation is called as **Skin passing** or **temper rolling**. It is a cold reduction method and the steel surface or skin is hardened by cold working, keeping the steel core soft & ductile. In fact, temper rolling does impart a small amount of cold reduction, typically between 0.25 and 1.0 percent.

The Skin Pass Mill where temper rolling is carried out is normally a single stand 4-high mill.

Both single and double stand Skin pass mills are present in BSL & RSP.

Following are the main advantages of Skin Passing.

- a. Imparts different surface finishes to the strip required for painting, coating enamelling etc.
- b. Gives a flat surface to the strip.
- c. Imparts the desired mechanical properties to the strip.
- d. Keeps the strip free from stretcher strains and luder bands that may develop during the forming operations.
- e. The flatness is improved, and the coil is oiled with rust preventive oil.

The skin passed coils are the packed and dispatched to stock yards or Customers as CR coils.

Sheet Sheering Line (SSL):

Some Coils are sheared into different lengths in Sheet Shearing lines and sent to Customers as CR sheets. SSL consists of one uncoiler & a Flying Shear to cut sheets of different lengths. Online inspection is done in most of the cases.

CR SLITTER

In slitter CR coils are slitted lengthwise and also to remove side trims to obtain uniform width throughout the coil as customer requirement.

CUT TO LENGTH LINE (CTL)

In CTL slitted coils are sheared to the desired length as per customer requirement.

Coated Sheet

SAIL's family of coated steel sheet products includes both hot-dipped and electrolytically-applied coatings. The protective coatings add superior corrosion resistance to the many other desirable properties of steel.

Electrolytic Tinning Line (ETL) Complex at RSP:

Here the Coating of tin is done by employing the principles of electrolysis in a acidic medium.

The continuous Electrolytic Tinning Line produces a shining tin coated surface in a variety of coating thickness. The tin plate shearing lines are equipped with sensitive pin hole detectors and an automatic off gauge detection system.

Continuous Galvanizing Lines at RSP and BSL:

Galvanizing Lines in both RSP & BSL are Sendzimer type Continuous Hot Dip Galvanizing facilities for On-line Oxidation Furnace for removing oil, grease, On-Line Reduction Furnace for annealing in protective atmosphere, Jet Coating for better control on Zinc coating thickness, Chemical Treatment to prevent atmospheric corrosion and Shearing facilities.

There are also multi-roller corrugating machines which produce corrugated sheets.

Shipping Section:

All Cold Rolled products like CR Coils/Sheets, ETP & GP/GC are packed, weighed, and dispatched through Road or Rail Wagons in Shipping Section.

6.9 Major Cold Rolling Defects

Holes, Scale Pits/Scabs, Scratches, Roll Imprints / Roll Marks, Coil Breaks, Orange peel effect, Wavy edge, Centre buckle, Pinch, Bluing or Oxidation, Water stain /quench marks.

6.10 Introduction to Pipe Plants and Silicon Steel Plant

PIPE PLANTS (PPs)

Rourkela Steel Plant has two Pipe making mills

ELECTRICAL RESISTANCE WELDING PIPE PLANT (ERWPP)

The ERWPP in RSP is modernized in the year 2005 with high frequency welder and is named as HFW pipe plant. It is designed to produce IS, ASTM and API grades pipes up to grade API-5L-X70 of diameter ranging from 8⁵/₈” to 18” (219.1 mm to 457.2mm) Outside diameter with wall thickness from 4.8 mm to 12.7 mm. The IS grade pipes are used for transportation of water, slurry, sewerage, ash etc. IS 4270 grades pipes produced at ERWPP are used for structural jobs, deep wells etc. ASTM grades pipes are also produced to cater few orders. **API 5L pipes are exclusively used** for transportation of gas and petroleum products.



Hot rolled coils (from Hot Strip Mill of RSP or from out side sources like BSL) **are the main input material** for ERW Pipe Plant. This input material is cold formed to a tubular shape by gradual deformation and then welded by the combination of heat and pressure.

SPIRAL WELDED PIPE PLANT(SWPP)

This mill meets the demand of handling bulk transportation of crude oil from shore to Refineries, slurry transportation, water supply and sewerage disposal to civil engineering pilings.. SW Pipe Plant has the capacity of producing both IS and API grade pipes in the range of 16" to 72" (406.4 mm to 1828.6 mm) outer diameter with wall thickness of 5.6 to 14.2 mm.

Hot rolled coils (from Hot Strip Mill of RSP or from out side sources like BSL) are the main input material for ERW Pipe Plant and SW pipe plant. This input material is formed to a tubular shape by gradual deformation and then welded by the combination of heat and pressure in ERW pipe plant .The HR coils are formed in spiral form and welded with filler material in SW pipe plant.



SILICON STEEL MILL



Electrical steel is an iron alloy which may have from zero to 6.5% silicon (Si:5Fe). Commercial alloys usually have silicon content up to 3.2% (higher concentrations usually provoke brittleness during cold rolling). Manganese and aluminum can be added up to 0.5%.

Silicon significantly increases the electrical resistivity of the steel, which decreases the induced eddy currents and narrows the hysteresis loop of the material, thus lowering the core loss. However, the grain structure hardens and embrittles the metal, which adversely affects the workability of the material, especially when rolling it. When alloying, the concentration levels of carbon, sulfur, oxygen and nitrogen must be kept low, as these elements indicate the presence of carbides, sulfides, oxides and nitrides. These compounds, even in particles as small as one micrometer in diameter, increase hysteresis losses while also decreasing magnetic permeability. The presence of carbon has a more detrimental effect than sulfur or oxygen. Carbon also causes magnetic aging when it slowly leaves the solid solution and precipitates as carbides, thus resulting in an increase in power loss over time. For these reasons, the carbon level is kept to 0.005% or lower. The carbon level can be reduced by annealing the steel in a decarburizing atmosphere, such as hydrogen.

The silicon mill Complex is designed to produce of Cold Rolled Non-Oriented electrical grade Silicon Steel of various sizes and grades in the form of Coils as well as Sheets.

6.11 Rolling of Special steels (Stainless Steel)

For rolling of Stainless slabs the following activities are carried out in sequence:

Coil Build-up Line [CBL]:

Bell Annealing Furnace [BAF]

Annealing, Shot blasting and Pickling

After annealing coils are shot blasted and pickled to remove the scale for further cold rolling. In order to remove the residual scale sticking to the surface, pickling process is carried out.

Strip Grinding

Coils which require repair grinding is processed in the line using coarse emery belt to remove the surface defects (slivers, scratches, minor scale etc) in full or in part depending on the severity and nature of the defects.

Sendzimir Mill (Z Mill)

Sendizmir Mill is used for rolling stainless steel at Salem. This mill is a 20-high mill having two work rolls supported by eighteen back-up rolls. Coolant oil is used during rolling which helps in strip cooling and also lubricating the various moving parts.

Skin Pass Mill (SPM)

Skin passing is done for stainless steel coils using Mirror polish rolls to improve the shape and have a bright surface and uniform thickness

Sheet Grinding and Polishing Line

Sheet grinding and polishing machine is used to produce special finishes and hairline finish on stainless steel sheets.
